

Problem Statement:

Hygiene and sanitation requirements have always been the most significant factor to stay healthy in human life. Proper cleaning and sanitation measures are one of the important ways to fight against various viruses. In recent years, the use of modern technology in sanitation has helped to fight against infectious diseases. One such modern technology is disinfection robots. Disinfection robots are increasingly used all around the world. Use of these disinfection robots reduces the risk of infection, cost of traditional cleaning and disinfection, and most importantly acquires confidence and security of Individuals.

Code:

```
// Define motor pins
const int motor1Pin = 2;
const int motor2Pin = 3;
const int motor3Pin = 4;
const int motor4Pin = 5;
// Define ultrasonic sensor pins
const int trigPin1 = 6;
const int echoPin1 = 7;
const int trigPin2 = 8;
const int echoPin2 = 9;
// Define distance threshold to stop motors (in centimeters)
const int stopThreshold = 10;
// Define motor speed (delay in milliseconds)
const int motorDelay = 300; // Adjust this value to change the speed
bool motorsStarted = false;
bool rotateInProgress = false; // Flag to track if rotate function is in progress
void setup() {
  // Initialize serial communication
  Serial.begin(9600);
  // Set motor pins as outputs
  pinMode(motor1Pin, OUTPUT);
  pinMode(motor2Pin, OUTPUT);
  pinMode(motor3Pin, OUTPUT);
  pinMode(motor4Pin, OUTPUT);
  // Set ultrasonic sensor pins
```

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pinMode(trigPin1, OUTPUT);
pinMode(echoPin1, INPUT);
pinMode(trigPin2, OUTPUT);
pinMode(echoPin2, INPUT);
}

void loop() {
  if (!rotateInProgress) { // Check if rotate function is not in progress
    // Check distance using ultrasonic sensor 1
    long duration1, distance1;
    digitalWrite(trigPin1, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin1, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin1, LOW);
    duration1 = pulseIn(echoPin1, HIGH);
    distance1 = duration1 * 0.034 / 2;
    // Check distance using ultrasonic sensor 2
    long duration2, distance2;
    digitalWrite(trigPin2, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin2, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin2, LOW);
    duration2 = pulseIn(echoPin2, HIGH);
    distance2 = duration2 * 0.034 / 2;
    // Print distances to serial monitor (optional)
    Serial.print("Distance 1: ");
    Serial.print(distance1);
    Serial.print(" cm, Distance 2: ");
    Serial.print(distance2);
    Serial.println(" cm");
    // Check if both sensor conditions are met
    if (distance1 > stopThreshold && distance2 <= stopThreshold) {
      if (!motorsStarted) {
        // Start motors
        startMotors();
        motorsStarted = true;
      }
    } else {
      if (motorsStarted) {
        // Stop motors if sensor conditions are not met
        stopMotors();
        motorsStarted = false;
        rotateInProgress = true; // Set flag to indicate rotate function
        // is in progress
        rotate(); // Call rotate function
      }
    }
  }
}

```

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}  
}  
// Add a small delay to prevent spamming the ultrasonic sensors  
delay(100);  
}  
// Function to stop motors  
void stopMotors() {  
  digitalWrite(motor1Pin, LOW);  
  digitalWrite(motor2Pin, LOW);  
  digitalWrite(motor3Pin, LOW);  
  digitalWrite(motor4Pin, LOW);  
}  
// Function to start motors  
void startMotors() {  
  digitalWrite(motor1Pin, HIGH);  
  digitalWrite(motor2Pin, LOW);  
  digitalWrite(motor3Pin, HIGH);  
  digitalWrite(motor4Pin, LOW);  
}  
void rotate() {  
  digitalWrite(motor1Pin, HIGH);  
  digitalWrite(motor2Pin, LOW);  
  digitalWrite(motor3Pin, LOW);  
  digitalWrite(motor4Pin, LOW);  
  delay(600);  
  rotateInProgress = false; // Set flag to indicate rotate function is Done
```

Photos:

